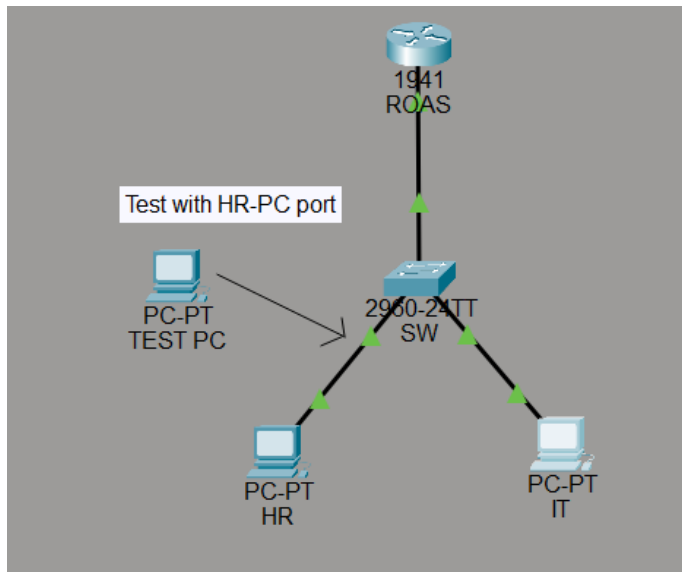


MINI PROJECT 11

Layer-2 Security (Port Security + DHCP Snooping + DAI) with Router as DHCP Server

DIAGRAM:



DESCRIPTION:

This project explains how Layer-2 security features like Port Security, DHCP Snooping, and Dynamic ARP Inspection (DAI) protect the network from attacks such as MAC flooding, rogue DHCP servers, and ARP spoofing.

STEP 1: Create VLANs, Access Ports & Trunk:

Create Vlan 10 and Vlan 20 on SW

Enable access port on SW for Vlan 10 and vlan 20

allow trunk on SW

STEP 2: Router as DHCP Server (ROAS):

```
interface g0/0
```

```
no shutdown
```

```
exit
```

```
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

```
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

STEP 3 : Configure Router as DHCP (Act as ROAS and DHCP):

```
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
```

```
ip dhcp pool HR
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
exit
```

```
ip dhcp pool IT
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
exit
```

STEP 4: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1

STEP 5: Enable DHCP Snooping, DAI(Dynamic Arp Inspection) & Port Security:

DHCP SNOOPING:

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
```

Note: trust only router uplink(do not trust acces ports)
interface fa0/24
ip dhcp snooping trust
exit

DAI:

ip arp inspection vlan 10,20

trust only router uplink same like dhcp snooping,

interface fa0/24
ip arp inspection trust
exit

Port security:

interface range fa0/1 - 2
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown (default mode)
exit

STEP 6 : Verification & Ping Test:

show ip dhcp snooping binding
show ip arp inspection statistics
show port-security interface fa0/1

connect test pc to fa0/1 verify port security is enabled.

Note: If you try to config pc with dhcp, it will fail(169.x.x.x) due to Discover packet modification by dhcp snooping, so disable the dhcp information option to assign,

no ip dhcp snooping information option

Ping test from IT PC to HR PC:

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms
```